WALL CLIMBER TREE

A catalogue of 25 grown trees

Trees that grow out of the wall, for climbing and sprawling plants. Every tree is grown branch by branch, bending and twisting as it goes, and every one is printed without supports.



The main design

Iteration 7: the tree the catalogue grows from. 285 × 142 × 183 mm, Cocoa Brown PLA. The 25 designs that follow each have their own crown and their own roots.

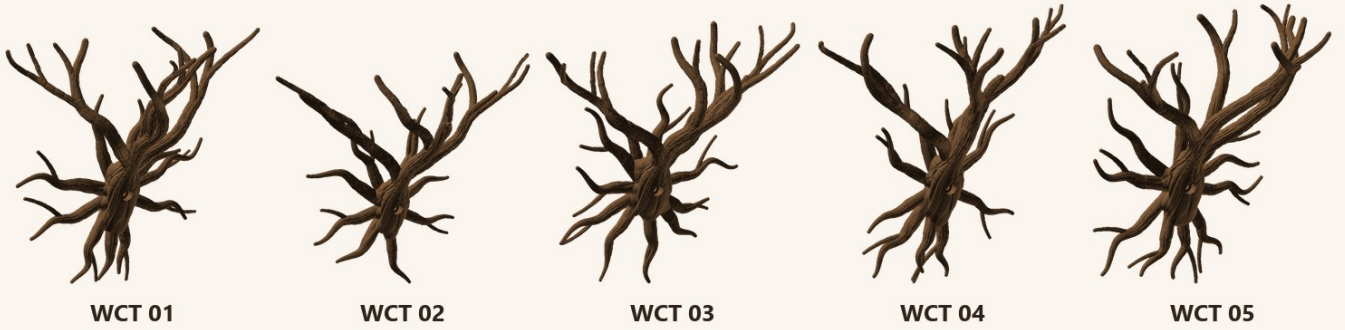
The collection

Three families, by how hard the branches bend. All share the same burl, slide and screw; crowns and roots are grown anew for each.

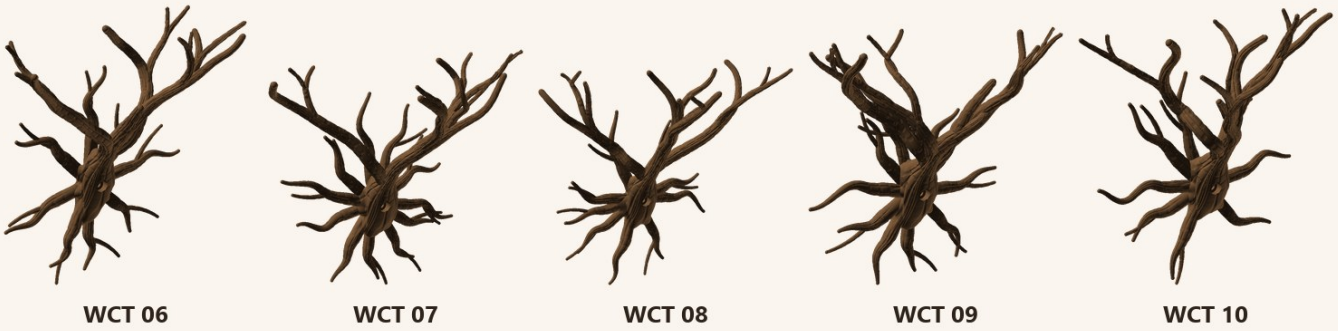
Grown · 01–09
bends like the main design

Swaying · 10–17
bends half as hard again

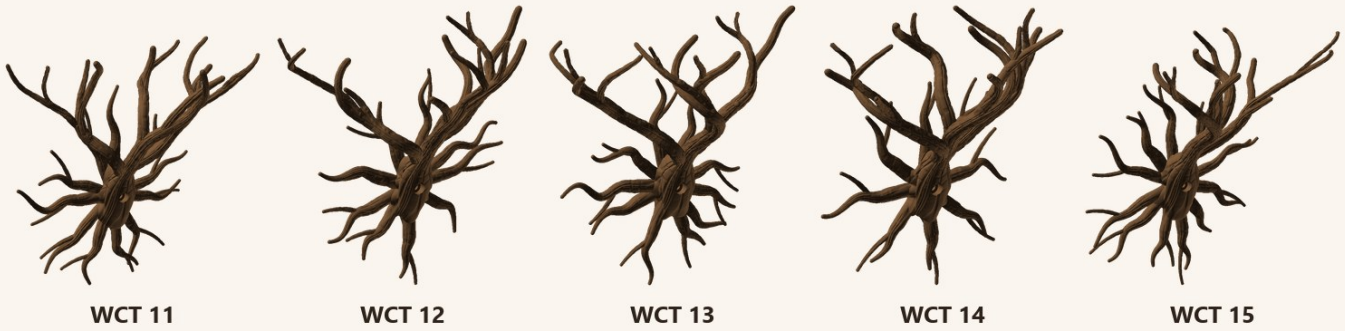
Wild · 18–25
bends and twists hardest



WCT 01 WCT 02 WCT 03 WCT 04 WCT 05



WCT 06 WCT 07 WCT 08 WCT 09 WCT 10



WCT 11 WCT 12 WCT 13 WCT 14 WCT 15



WCT 16 WCT 17 WCT 18 WCT 19 WCT 20



WCT 21 WCT 22 WCT 23 WCT 24 WCT 25



WCT 01

Grown · ×1 bend

Branches turn
143° on average

Back to the wall
257 mm of branch

Tips span
258 mm along the wall

Roots
10, meander 6.5 mm

Tree
262.7 × 140.3 × 181.6 mm

Print, tree
3h 38m, 70 g

Print, roots
1h 54m, 56 g

Seeds
crown 7010 · roots 101

Print-checked



WCT 02

Grown · ×1 bend

Branches turn
141° on average

Back to the wall
146 mm of branch

Tips span
278 mm along the wall

Roots
8, meander 6.5 mm

Tree
281.0 × 139.4 × 181.7 mm

Print, tree
3h 31m, 68 g

Print, roots
1h 31m, 46 g

Seeds
crown 7020 · roots 102

Print-checked



WCT 03

Grown · ×1 bend

Branches turn
155° on average

Back to the wall
186 mm of branch

Tips span
281 mm along the wall

Roots
11, meander 6.5 mm

Tree
285.1 × 142.5 × 180.9 mm

Print, tree
3h 47m, 72 g

Print, roots
2h 0m, 59 g

Seeds
crown 7030 · roots 103

Print-checked



WCT 04

Grown · ×1 bend

Branches turn
141° on average

Back to the wall
131 mm of branch

Tips span
279 mm along the wall

Roots
9, meander 6.5 mm

Tree
283.4 × 129.5 × 182.5 mm

Print, tree
3h 27m, 67 g

Print, roots
1h 44m, 51 g

Seeds
crown 7040 · roots 104

Print-checked



WCT 05

Grown · ×1 bend

Branches turn
147° on average

Back to the wall
315 mm of branch

Tips span
235 mm along the wall

Roots
10, meander 6.5 mm

Tree
239.3 × 142.9 × 179.1 mm

Print, tree
3h 40m, 71 g

Print, roots
1h 52m, 55 g

Seeds
crown 7050 · roots 105

Print-checked



WCT 06

Grown · ×1 bend

Branches turn
115° on average

Back to the wall
155 mm of branch

Tips span
249 mm along the wall

Roots
8, meander 6.5 mm

Tree
253.7 × 141.3 × 182.6 mm

Print, tree
3h 32m, 69 g

Print, roots
1h 35m, 48 g

Seeds
crown 7060 · roots 106

Print-checked



WCT 07

Grown · ×1 bend

Branches turn
138° on average

Back to the wall
87 mm of branch

Tips span
279 mm along the wall

Roots
11, meander 6.5 mm

Tree
284.9 × 143.9 × 180.6 mm

Print, tree
3h 46m, 74 g

Print, roots
2h 1m, 60 g

Seeds
crown 7070 · roots 107

Print-checked



WCT 08

Grown · ×1 bend

Branches turn
136° on average

Back to the wall
172 mm of branch

Tips span
254 mm along the wall

Roots
9, meander 6.5 mm

Tree
258.6 × 142.6 × 180.2 mm

Print, tree
3h 35m, 70 g

Print, roots
1h 41m, 50 g

Seeds
crown 7080 · roots 108

Print-checked



WCT 09

Grown · ×1 bend

Branches turn
149° on average

Back to the wall
181 mm of branch

Tips span
238 mm along the wall

Roots
10, meander 6.5 mm

Tree
240.7 × 140.5 × 181.8 mm

Print, tree
3h 34m, 70 g

Print, roots
1h 54m, 57 g

Seeds
crown 7090 · roots 109

Print-checked



WCT 10

Swaying · ×1.5 bend

Branches turn
155° on average

Back to the wall
95 mm of branch

Tips span
273 mm along the wall

Roots
8, meander 7 mm

Tree
276.8 × 141.6 × 180.5 mm

Print, tree
3h 34m, 70 g

Print, roots
1h 35m, 47 g

Seeds
crown 7100 · roots 110

Print-checked



WCT 11

Swaying · ×1.5 bend

Branches turn
168° on average

Back to the wall
212 mm of branch

Tips span
255 mm along the wall

Roots
11, meander 7 mm

Tree
259.8 × 139.0 × 178.4 mm

Print, tree
3h 49m, 73 g

Print, roots
2h 4m, 61 g

Seeds
crown 7110 · roots 111

Print-checked



WCT 12

Swaying · ×1.5 bend

Branches turn
171° on average

Back to the wall
215 mm of branch

Tips span
272 mm along the wall

Roots
9, meander 7 mm

Tree
276.7 × 141.1 × 182.8 mm

Print, tree
3h 44m, 73 g

Print, roots
1h 44m, 52 g

Seeds
crown 7120 · roots 112

Print-checked



WCT 13

Swaying · ×1.5 bend

Branches turn
181° on average

Back to the wall
264 mm of branch

Tips span
285 mm along the wall

Roots
10, meander 7 mm

Tree
291.3 × 139.9 × 181.8 mm

Print, tree
3h 57m, 78 g

Print, roots
1h 50m, 54 g

Seeds
crown 7130 · roots 113

Print-checked



WCT 14

Swaying · ×1.5 bend

Branches turn
176° on average

Back to the wall
314 mm of branch

Tips span
259 mm along the wall

Roots
8, meander 7 mm

Tree
263.5 × 140.6 × 182.3 mm

Print, tree
3h 52m, 74 g

Print, roots
1h 36m, 47 g

Seeds
crown 7140 · roots 114

Print-checked



WCT 15

Swaying · ×1.5 bend

Branches turn
168° on average

Back to the wall
196 mm of branch

Tips span
236 mm along the wall

Roots
11, meander 7 mm

Tree
240.6 × 141.9 × 179.9 mm

Print, tree
3h 27m, 68 g

Print, roots
2h 7m, 62 g

Seeds
crown 7150 · roots 115

Print-checked



WCT 16

Swaying · ×1.5 bend

Branches turn
149° on average

Back to the wall
130 mm of branch

Tips span
274 mm along the wall

Roots
9, meander 7 mm

Tree
280.2 × 141.6 × 183.5 mm

Print, tree
3h 40m, 72 g

Print, roots
1h 40m, 50 g

Seeds
crown 7160 · roots 116

Print-checked



WCT 17

Swaying · ×1.5 bend

Branches turn
156° on average

Back to the wall
141 mm of branch

Tips span
222 mm along the wall

Roots
10, meander 7 mm

Tree
225.9 × 142.3 × 179.4 mm

Print, tree
3h 43m, 72 g

Print, roots
1h 54m, 57 g

Seeds
crown 7170 · roots 117

Print-checked



WCT 18

Wild · ×2.2 bend

Branches turn
229° on average

Back to the wall
236 mm of branch

Tips span
200 mm along the wall

Roots
8, meander 8 mm

Tree
203.1 × 142.8 × 180.2 mm

Print, tree
3h 35m, 69 g

Print, roots
1h 36m, 48 g

Seeds
crown 7180 · roots 118

Print-checked



WCT 19

Wild · ×2.2 bend

Branches turn
215° on average

Back to the wall
235 mm of branch

Tips span
266 mm along the wall

Roots
11, meander 8 mm

Tree
269.7 × 144.5 × 182.3 mm

Print, tree
3h 50m, 75 g

Print, roots
2h 1m, 61 g

Seeds
crown 7190 · roots 119

Print-checked



WCT 20

Wild · ×2.2 bend

Branches turn
204° on average

Back to the wall
283 mm of branch

Tips span
247 mm along the wall

Roots
9, meander 8 mm

Tree
251.1 × 128.9 × 178.9 mm

Print, tree
3h 46m, 73 g

Print, roots
1h 46m, 52 g

Seeds
crown 7200 · roots 120

Print-checked



WCT 21

Wild · ×2.2 bend

Branches turn
203° on average

Back to the wall
241 mm of branch

Tips span
245 mm along the wall

Roots
10, meander 8 mm

Tree
248.4 × 138.3 × 179.9 mm

Print, tree
3h 43m, 72 g

Print, roots
1h 45m, 52 g

Seeds
crown 7210 · roots 121

Print-checked



WCT 22

Wild · ×2.2 bend

Branches turn
187° on average

Back to the wall
325 mm of branch

Tips span
273 mm along the wall

Roots
8, meander 8 mm

Tree
275.9 × 141.1 × 182.8 mm

Print, tree
3h 45m, 72 g

Print, roots
1h 32m, 47 g

Seeds
crown 7220 · roots 122

Print-checked



WCT 23

Wild · ×2.2 bend

Branches turn
218° on average

Back to the wall
229 mm of branch

Tips span
256 mm along the wall

Roots
11, meander 8 mm

Tree
260.4 × 122.3 × 182.0 mm

Print, tree
3h 55m, 76 g

Print, roots
2h 2m, 61 g

Seeds
crown 7230 · roots 123

Print-checked



WCT 24

Wild · ×2.2 bend

Branches turn
183° on average

Back to the wall
258 mm of branch

Tips span
222 mm along the wall

Roots
9, meander 8 mm

Tree
225.7 × 131.3 × 181.0 mm

Print, tree
3h 33m, 68 g

Print, roots
1h 52m, 55 g

Seeds
crown 7240 · roots 124

Print-checked



WCT 25

Wild · ×2.2 bend

Branches turn
205° on average

Back to the wall
144 mm of branch

Tips span
232 mm along the wall

Roots
10, meander 8 mm

Tree
236.1 × 139.9 × 180.1 mm

Print, tree
3h 46m, 73 g

Print, roots
1h 54m, 56 g

Seeds
crown 7250 · roots 125

Print-checked

Printing and hanging



Two parts

The roots print flat on their back. The tree prints standing, just as it hangs, with no supports: every branch climbs at least 32°, inside the 60° overhang limit. Use a brim; the tree stands on a 4 cm^2 foot.

Material

Bambu PLA Basic, Cocoa Brown #6F5034. Bambu Lab H2C, 0.4 mm nozzle, 0.20 mm Standard, 35 % sparse infill. The widest trees need the full bed.

Hanging

1. Hold the roots against the wall. 2. Slide the tree down the dovetail until its foot sits on the burl's shelf. 3. Drive one 6 mm screw (6 × 50 mm or longer, with a wall plug) through the knot hole under the trunk: the tree clamps the roots to the wall.

Every design is checked

Both parts: one closed solid, overhang under 0.5 % of the surface, no region starting in mid-air, a clean Bambu Studio slice, and a slide that fits with 0 mm^3 of overlap all the way in.